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ECEN 380
3/9/2026

PS09: 3.61(b), 3.62(a), 3.68(a)(c), 3.70(a)(d), MATLAB 3.117

3.61(b) Use the table of transforms and properties to find the DTFT of the following signal:

$$x[n] = (n-2)(u[n+4]-u[n-5])$$

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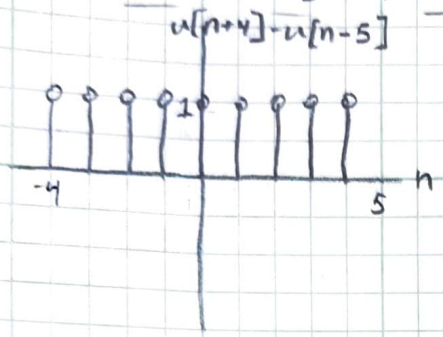

$$\Rightarrow \begin{cases} 1, & -4 \le n \le 4 \\ 0, & \text{otherwise} \end{cases}$$

$$x[n] = (n-2) \cdot \begin{cases} 1, & |n| \le 4 \\ 0, & \text{otherwise} \end{cases}$$

C.7 : $-jn\,x[n] \xleftrightarrow{DTFT} \frac{d}{d\Omega} X(e^{j\Omega})$

$$(n)\,x[n] \xleftrightarrow{DTFT} \frac{d}{d\Omega} X(e^{j\Omega}) \cdot \frac{1}{-j} = j\frac{d}{d\Omega} X(e^{j\Omega})$$

C.3 : $x[n] = \begin{cases} 1, & |n| \le M \\ 0, & \text{other} \end{cases} \xleftrightarrow{DTFT} X(e^{j\Omega}) = \dfrac{\sin\left[\Omega\left(\frac{2M+1}{2}\right)\right]}{\sin\left(\frac{\Omega}{2}\right)}$

C.7 : $x[n-n_0] \xleftrightarrow{DTFT} e^{-j\Omega n_0} X(e^{j\Omega})$

C.7 : $ax[n] + by[n] \xleftrightarrow{DTFT} aX(e^{j\Omega}) + bY(e^{j\Omega})$

$$x[n] = n \cdot \begin{cases} 1, & |n| \le 4 \\ 0, & \text{otherwise} \end{cases} - 2 \cdot \begin{cases} 1, & |n| \le 4 \\ 0, & \text{otherwise} \end{cases} \quad \text{w}[n]$$

$$W(e^{j\Omega}) = \frac{\sin\left(\frac{9}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}$$

$$\text{DTFT}\{n\,w[n]\} = j\frac{d}{d\Omega}\frac{\sin\left(\frac{9}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}$$

$$\text{DTFT}\{-2w[n]\} = -2\,\frac{\sin\left(\frac{9}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}$$

$$\boxed{\text{DTFT}\{x[n]\} = X(e^{j\Omega}) = j\frac{d}{d\Omega}\left[\frac{\sin\left(\frac{9}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}\right] - 2 \cdot \frac{\sin\left(\frac{9}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}}$$

3.62(a) Use the FT pair:

$$x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases} \xleftrightarrow{FT} X(j\omega) = \frac{2\sin(\omega)}{\omega}$$

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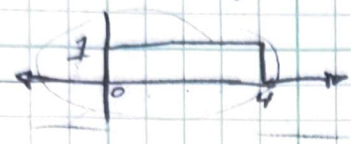
and the FT properties to evaluate the frequency-domain representation of the signal depicted:



$$y(t) = x(\tfrac{1}{4}t), \text{ where } x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

C.7: $x(at) \xleftarrow{FT} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$

$$X(j\omega) = \frac{2\sin(\omega)}{\omega} \Longrightarrow FT\{x(\tfrac{1}{4}t)\} = \frac{1}{|\tfrac{1}{4}|} \cdot \frac{2\sin(\omega/\tfrac{1}{4})}{\omega/\tfrac{1}{4}}$$

$$Y(j\omega) = 4 \cdot \frac{2\sin(4\omega)}{4\omega} = \frac{2\sin(4\omega)}{\omega}$$

$$\boxed{Y(j\omega) = \frac{2\sin(4\omega)}{\omega}}$$

3.68 (a) Determine the frequency and the impulse response for the system described by the following differential equation:

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(a) $\frac{d}{dt} y(t) + 3y(t) = x(t)$

Class slides:

$FT\{h(t)\} = H(j\omega)$; $y(t) = h(t) * x(t)$

impulse response — freq. response

$Y(j\omega) = H(j\omega) X(j\omega)$

$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$

C.7: $\frac{d}{dt} x(t) \xleftrightarrow{FT} j\omega X(j\omega)$

$$(j\omega) Y(j\omega) + 3Y(j\omega) = X(j\omega)$$

$$(j\omega + 3) Y(j\omega) = X(j\omega)$$

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{1}{j\omega + 3} = H(j\omega)$$

C.4: $x(t) = e^{-at} u(t)$, $Re\{a\} > 0 \xleftrightarrow{FT} X(j\omega) = \frac{1}{a + j\omega}$

$$Inv.FT\{H(j\omega)\} = Inv.FT\left\{\frac{1}{3 + j\omega}\right\} = e^{-3t} u(t) = h(t)$$

Impulse response : $h(t) = e^{-3t} u(t)$

Frequency response : $H(j\omega) = \frac{1}{j\omega + 3}$

3.68(c) Determine frequency & impulse responses for the system described by the following difference equation:

(c) $y[n] - \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = 3x[n] - \frac{3}{4}x[n-1]$

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$DTFT\{h[n]\} = H(e^{j\Omega})$; $y[n] = h[n] * x[n]$

$Y(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\omega})$

$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$

C.7 : $x[n-n_0] \xleftrightarrow{DTFT} e^{-j\Omega n_0}X(e^{j\Omega})$

$$Y(e^{j\Omega}) - \frac{1}{4}e^{-j\Omega(1)}Y(e^{j\Omega}) - \frac{1}{8}e^{-j\Omega(2)}Y(e^{j\Omega}) = 3X(e^{j\Omega}) - \frac{3}{4}e^{-j\Omega(1)}X(e^{j\Omega})$$

$$Y(e^{j\Omega})\left[1 - \frac{1}{4}e^{-j\Omega} - \frac{1}{8}e^{-2j\Omega}\right] = X(e^{j\Omega})\left[3 - \frac{3}{4}e^{-j\Omega}\right]$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{3 - \frac{3}{4}e^{-j\Omega}}{1 - \frac{1}{4}e^{-j\Omega} - \frac{1}{8}e^{-2j\Omega}} = \frac{3 - \frac{3}{4}e^{-j\Omega}}{-\frac{1}{8}(e^{-j\Omega}+4)(e^{-j\Omega}-2)}$$

$-\frac{1}{8}x^2 - \frac{1}{4}x + 1$

$-\frac{1}{8}(x^2 + 2x - 8)$

$-\frac{1}{8}(x+4)(x-2)$ ($x = e^{-j\Omega}$ for now)

$$= \frac{-24 + 6e^{-j\Omega}}{(e^{-j\Omega}+4)(e^{-j\Omega}-2)} = \frac{A}{(e^{-j\Omega}+4)} + \frac{B}{(e^{-j\Omega}-2)}$$

$$\frac{-24 + 6x}{(x+4)(x-2)} = \frac{A}{(x+4)} + \frac{B}{(x-2)} \Rightarrow -24 + 6x = A(x-2) + B(x+4)$$

$$6x + -24 = Ax - 2A + Bx + 4B$$

$A + B = 6$

$-2A + 4B = -24$

$$\begin{bmatrix} 1 & 1 & 6 \\ -2 & 4 & -24 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= \frac{8}{(e^{-j\Omega}+4)} + \frac{-2}{(e^{-j\Omega}-2)}$$

C.3 : $x(e^{j\Omega}) = \frac{1}{1-\alpha e^{-j\Omega}} \xleftrightarrow{DTFT} x[n]\alpha^n u[n], |\alpha|<1$

$$H(e^{j\omega}) = \frac{8}{(e^{-j\omega}+4)} + \frac{-2}{(e^{-j\omega}-2)} ; \quad \frac{8}{4(1+\frac{1}{4}e^{-j\omega})} = \frac{8}{4(1-(-\frac{1}{4}e^{-j\omega}))} \quad \boxed{5/6}$$

$$= \frac{2}{1-(-\frac{1}{4}e^{-j\omega})}$$

$$\frac{-2}{-2(1-\frac{1}{2}e^{-j\omega})} = \frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

$$H(e^{j\omega}) = \underbrace{\frac{2}{1-(-\frac{1}{4}e^{-j\omega})}} + \frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

$$h[n] = \text{Inv. DTFT}\{\ \} = 2\left[-\tfrac{1}{4}^{n} u[n]\right] + \tfrac{1}{2}^{n} u[n]$$

$$\boxed{\begin{aligned} H(e^{j\omega}) &= \frac{2}{1-(-\frac{1}{4}e^{-j\omega})} + \frac{1}{1-\frac{1}{2}e^{-j\omega}} \quad \text{(frequency response)} \\ h[n] &= 2\left(-\tfrac{1}{4}\right)^{n} u[n] + \left(\tfrac{1}{2}\right)^{n} u[n] \quad \text{(impulse response)} \end{aligned}}$$

3.70(a) Determine differential-equation description for system w/ frequency response:

$$H(j\omega) = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$Y(j\omega)(1 + 2j\omega) = X(j\omega)(2 + 3j\omega - 3(j\omega)^2)$$

$$Y(j\omega) + 2j\omega Y(j\omega) = 2X(j\omega) + 3j\omega X(j\omega) - 3(j\omega)^2 X(j\omega)$$

$$\boxed{y(t) + 2\frac{d}{dt}y(t) = 2x(t) + 3\frac{d}{dt}x(t) - 3\frac{d^2}{dt^2}x(t)}$$

3.70(d) Determine difference-equation description for system w/ frequency response:

$$H(e^{j\Omega}) = \frac{1 + e^{-j\Omega}}{e^{-j2\Omega} + 3}$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1 + e^{-j\Omega}}{e^{-j2\Omega} + 3}$$

$$Y(e^{j\Omega})(e^{-j2\Omega} + 3) = X(e^{j\Omega})(1 + e^{-j\Omega})$$

$$e^{-j2\Omega} Y(e^{j\Omega}) + 3Y(e^{j\Omega}) = X(e^{j\Omega}) + e^{-j\Omega} X(e^{j\Omega})$$

$$\boxed{y[n-2] + 3y[n] = x[n] + x[n-1]}$$